Conditional Probability and Bayes Rule

ANSWER KEY

Section 1: Joint Probability and Independence

1 If two events A and B are statistically independent with $P(A) = 0.4$ and $P(B) = 0.3$, find the joint probability $P(A \cap B)$.

a. 0.12 ✓

b. 0.70

c. 0.10

d. 0.30

By independence: $P(A \cap B) = P(A) \times P(B) = 0.4 \times 0.3 = 0.12$.

2 Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, find $P(A \cap B)$.

a. 0.20 ✓

b. 0.10

c. 0.30

d. 0.15

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.7 = 0.20$.

3 Are the events in question 2 statistically independent?

a. **Yes, because** $P(A)P(B) = 0.5 \times 0.4 = 0.20 = P(A \cap B)$ ✓

b. No, because $P(A) \neq P(B)$

c. No, because $P(A \cup B) \neq 1$

d. Cannot be determined without conditional probabilities

Since $P(A \cap B) = 0.20 = P(A)P(B)$, the definition of independence is satisfied exactly.

Section 2: Conditional Probability and the Chain Rule

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4 Given $P(A) = 0.6$ and $P(A \cap B) = 0.18$, calculate the conditional probability $P(B|A)$.

a. 0.30 ✓

b. 0.108

c. 0.42

d. 0.25

$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.60} = 0.30$.

5 In an autoregressive language model, write the joint probability of a sequence of three tokens $P(w_1, w_2, w_3)$ using the probability Chain Rule.

a. $P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)$ ✓

b. $P(w_1) + P(w_2) + P(w_3)$

c. $P(w_1)P(w_2)P(w_3)$

d. $P(w_3|w_2)P(w_2|w_1)$

By exact probability decomposition: $P(w_1, w_2, w_3) = P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)$.

6 If $P(A|B) = 0.8$ and $P(B) = 0.25$, find $P(A \cap B)$.

a. 0.20 ✓

b. 0.55

c. 0.3125

d. 0.10

$P(A \cap B) = P(A|B)P(B) = 0.8 \times 0.25 = 0.20$.

Section 3: Law of Total Probability

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7 Factory line 1 produces 60% of microchips with defect rate 2%. Line 2 produces 40% with defect rate 5%. What is the total probability that a random chip is defective?

a. 0.032 ✓

b. 0.070

c. 0.035

d. 0.025

$P(D) = P(D|L_1)P(L_1) + P(D|L_2)P(L_2) = (0.02)(0.60) + (0.05)(0.40) = 0.012 + 0.020 = 0.032.$

8 In question 7, if a selected chip is defective, what is the probability it came from Line 2?

a. $\frac{0.020}{0.032} = 0.625$ ✓

b. 0.400

c. 0.375

d. 0.500

Bayes: $P(L_2|D) = \frac{P(D|L_2)P(L_2)}{P(D)} = \frac{0.020}{0.032} = 0.625.$

9 State the Law of Total Probability for an exhaustive partition $\{A_1, A_2, \ldots, A_k\}$ of the sample space.

a. $P(B) = \sum_{i=1}^{k} P(B|A_i)P(A_i)$ ✓

b. $P(B) = \prod_{i=1}^{k} P(B|A_i)$

c. $P(B) = \max_i P(B|A_i)$

d. $P(B) = \sum_{i=1}^{k} P(A_i|B)$

Every element in B belongs to exactly one partition A_i, giving $P(B) = \sum P(B \cap A_i) = \sum P(B|A_i)P(A_i)$.

Section 4: Bayes Theorem in Machine Learning and Screening

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10 In medical screening, a disease has prevalence $P(D) = 0.01$. The test has sensitivity $P(+|D) = 0.99$ and false positive rate $P(+|\neg D) = 0.02$. What is the posterior probability $P(D|+)$ that a positive patient has the disease?

a. $\approx 33.3\%$ ✓

b. 99.0%

c. 50.0%

d. 5.0%

$P(+) = (0.99)(0.01) + (0.02)(0.99) = 0.0099 + 0.0198 = 0.0297$. $P(D|+) = \frac{0.0099}{0.0297} = \frac{1}{3} \approx 33.3\%$.

11 What is the foundational simplifying assumption of the Naive Bayes text classifier?

a. **Features (words) are conditionally independent given the class label** ✓

b. All prior probabilities are equal

c. The vocabulary size is infinite

d. Features follow a uniform distribution

Naive Bayes assumes $P(x_1, \ldots, x_n|C) = \prod_{i=1}^{n} P(x_i|C)$, dramatically reducing parameter count.

12 In Maximum A Posteriori (MAP) estimation, how does the optimization objective differ from standard Maximum Likelihood Estimation (MLE)?

a. **MAP incorporates a prior distribution:** $\theta_{MAP} = \arg\max[\ln P(D|\theta) + \ln P(\theta)]$ ✓

b. MAP ignores all observed training data

c. MAP only applies to linear models

d. MAP minimizes variance instead of error

MAP maximizes posterior $\propto$ likelihood $\times$ prior, functioning mathematically like regularized MLE.

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